Department of Artificial Intelligence

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Concentration Inequalities (AI2200)

Lecture 04

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Application 2: Classification

- Let $\mathcal{X}$ be a generic feature space and $\mathcal{Y}$ be a finite label space with $|\mathcal{Y}| = K$
- Let $P_{X,Y}$ be an unknown probability distribution on $\mathcal{X} \times \mathcal{Y}$
- Let $g : \mathcal{X} \to \mathcal{Y}$ be a classifier mapping $\mathcal{X}$ to $\mathcal{Y}$
- The performance of classifier g is measured by its risk:

$$R(g) \;:=\; \mathbb{P}_{(X,Y)\sim P_{X,Y}}\Big(\{g(X) \neq Y\}\Big).$$

- Let $\mathcal{G}$ be a finite set of classifiers, and let

$$g^\star \;=\; \arg\min_{g \in \mathcal{G}} R(g)$$

  be the optimal classifier in $\mathcal{G}$
- Because $P_{X,Y}$ is unknown, it is not possible to identify $g^\star$ explicitly

Important Question

Without knowing $P_{X,Y}$, can we build a classifier that can closely compete with $g^\star$? **Yes, with data!**

Example: Binary Classification

Which images contain a car?

Pic credits: Codex + GPT 5.6 Sol

Training data

Image	Label
X_1	$Y_1 = 1$
X_2	$Y_2 = 0$
X_3	$Y_3 = 1$
$\vdots$	$\vdots$
X_n	$Y_n = 1$

1 = car   0 = no car

$$(X_1, Y_1), \ldots, (X_n, Y_n) \overset{\text{IID}}{\sim} P_{X,Y}$$

Training Data and Empirical Risk Minimisation (ERM)

- Suppose we take n training data points $(X_1, Y_1), \ldots, (X_n, Y_n) \overset{\text{IID}}{\sim} P_{X,Y}$,

$P_{X,Y}$ unknown

- We want to build a classifier using the data $\mathcal{D}_n = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$
- One such classifier is the so-called **empirical risk minimiser**:

$$\widehat{R}_n(g) := \underbrace{\frac{1}{n} \sum_{k=1}^{n} \mathbf{1}_{\{g(X_k) \neq Y_k\}}}_{\text{empirical risk of } g}, \qquad \widehat{g}_n := \arg\min_{g \in \mathcal{G}} \widehat{R}_n(g).$$

- The empirical risk minimiser, $\widehat{g}_n$, depends on the data

Important Question

How well does the empirical risk minimiser $\widehat{g}_n$ perform relative to the optimal classifier $g^\star$?

Risk of the Empirical Risk Minimiser $\widehat{g}_n$

- For any $g \in \mathcal{G}$ that is not based on the data $\mathcal{D}_n$,

$$\mathbb{E}[\widehat{R}_n(g)] = \mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{g(X_k) \neq Y_k\}}\right] = \frac{1}{n}\sum_{k=1}^{n} \mathbb{E}[\mathbf{1}_{\{g(X_k) \neq Y_k\}}] = \frac{1}{n}\sum_{k=1}^{n} \mathbb{P}(\{g(X_k) \neq Y_k\})$$

$$= \mathbb{P}(\{g(X_1) \neq Y_1\}) \qquad (X_1, Y_1), \ldots, (X_n, Y_n) \text{ are identically distributed}$$

$$= \mathbb{P}_{(X,Y)\sim P_{X,Y}}(\{g(X) \neq Y\}) \qquad (X_1, Y_1) \sim P_{X,Y} \qquad = R(g).$$

- When we repeat the above steps for $g = \widehat{g}_n$ (which is based on $\mathcal{D}_n$), we cannot go from third line to fourth line:

$$\frac{1}{n}\sum_{k=1}^{n} \mathbb{P}(\{\widehat{g}_n(X_k) \neq Y_k\}) \neq \mathbb{P}(\{\widehat{g}_n(X_1) \neq Y_1\}).$$

- The quantity

$$R(\widehat{g}_n) = \mathbb{P}_{(X,Y)\sim P_{X,Y}}\Big(\{\widehat{g}_n(X) \neq Y\}\Big),$$

is a random variable as it depends on the data $\mathcal{D}_n$!

Bernoulli Concentration

- For any $g \in \mathcal{G}$ that does not depend on the data $\mathcal{D}_n$, and for any $\varepsilon > 0$,

$$\begin{aligned}
\mathbb{P}\left(\left\{\left|\widehat{R}_n(g) - R(g)\right| \geq \varepsilon\right\}\right) &\leq \mathbb{P}\left(\left\{\left|\widehat{R}_n(g) - R(g)\right| \geq \varepsilon R(g)\right\}\right) && (R(g) \leq 1) \\
&\leq 2\exp\left(-\frac{n\varepsilon^2 R^2(g)}{2+\varepsilon}\right) \\
&\leq 2\exp\left(-\frac{n\varepsilon^2 R^2(g^\star)}{2+\varepsilon}\right) && (R(g^\star) \leq R(g) \text{ for all } g \in \mathcal{G} \text{ not depending on } \mathcal{D}_n).
\end{aligned}$$

- We cannot apply the above result to upper bound

$$\mathbb{P}\left(\left\{\left|\widehat{R}_n(\widehat{g}_n) - R(\widehat{g}_n)\right| \geq \varepsilon\right\}\right)$$

as $R(\widehat{g}_n)$ is a random quantity!

Performance of $\widehat{g}_n$ Relative to $g^\star$

- Notice that

$$\begin{aligned}\bigcap_{g\in\mathcal{G}} \left\{|\widehat{R}_n(g) - R(g)| < \varepsilon\right\} &\subseteq \left\{|\widehat{R}_n(\widehat{g}_n) - R(\widehat{g}_n)| < \varepsilon, \quad |\widehat{R}_n(g^\star) - R(g^\star)| < \varepsilon\right\} \\ &\subseteq \left\{R(\widehat{g}_n) < \widehat{R}_n(\widehat{g}_n) + \varepsilon, \quad \widehat{R}_n(g^\star) < R(g^\star) + \varepsilon\right\} \\ &\subseteq \left\{R(\widehat{g}_n) < R(g^\star) + 2\varepsilon\right\} \qquad \left(\text{using} \quad \widehat{R}_n(\widehat{g}_n) \leq \widehat{R}_n(g^\star)\right).\end{aligned}$$

- Thus, it follows that for any $\varepsilon > 0$,

$$\begin{aligned}\mathbb{P}\Big(\{R(\widehat{g}_n) \geq R(g^\star) + 2\varepsilon\}\Big) &\leq \mathbb{P}\left(\bigcup_{g\in\mathcal{G}} \{|\widehat{R}_n(g) - R(g)| \geq \varepsilon\}\right) \leq \sum_{g\in\mathcal{G}} \mathbb{P}\Big(\{|\widehat{R}_n(g) - R(g)| \geq \varepsilon\}\Big) \\ &\leq 2\,|\mathcal{G}|\,\exp\left(-\frac{n\varepsilon^2 R^2(g^\star)}{2+\varepsilon}\right).\end{aligned}$$

- Setting the right-hand side to δ, we see that

$$n \geq \left(\frac{2+\varepsilon}{\varepsilon^2\, R^2(g^\star)}\right) \log\left(\frac{2|\mathcal{G}|}{\delta}\right) \qquad \Longrightarrow \qquad \mathbb{P}\Big(\{R(\widehat{g}_n) \geq R(g^\star) + 2\varepsilon\}\Big) \leq \delta.$$

Empirical Risk Minimisation in a Nutshell

- Empirical risk minimisation is a learning algorithm that:
  - Takes as input n IID data points $(X_1, Y_1), \ldots, (X_n, Y_n)$.
  - Outputs a classifier $\widehat{g}_n$ that satisfies:

$$n \;\geq\; \left(\frac{2+\varepsilon}{\varepsilon^2\, R^2(g^\star)}\right) \log\left(\frac{2|\mathcal{G}|}{\delta}\right) \qquad \Longrightarrow \qquad \mathbb{P}\Big(\big\{R(\widehat{g}_n) \geq R(g^\star) + 2\varepsilon\big\}\Big) \;\leq\; \delta.$$

Probably Approximately Correct (PAC) Learning

Definition ((ε, δ)-PAC Property)

Fix a feature space $\mathcal{X}$ and a label space $\mathcal{Y}$. Let $\mathcal{G}$ be a finite collection of classifiers mapping $\mathcal{X}$ to $\mathcal{Y}$.
Let $R : \mathcal{G} \to [0, +\infty)$ denote a risk function, and let

$$g^\star = \arg\min_{g \in \mathcal{G}} R(g)$$

be the optimal classifier in $\mathcal{G}$.

- Given $\varepsilon > 0$ and $\delta \in (0, 1)$, we say that a classifier $g \in \mathcal{G}$ is (ε, δ)-PAC if:

$$\mathbb{P}\left(\left\{R(g) \geq R(g^\star) + \varepsilon\right\}\right) \leq \delta.$$

- The collection $\mathcal{G}$ is said to be PAC-learnable if:
for any choice of $\varepsilon > 0$ and $\delta \in (0, 1)$, there exists a positive integer $N_{\varepsilon,\delta,\mathcal{G}}$ and a learning algorithm that takes as input n IID data points $(X_1, Y_1), \ldots, (X_n, Y_n)$ and outputs a classifier $\widehat{g}_n$ satisfying

$$n \geq N_{\varepsilon,\delta,\mathcal{G}} \quad \Longrightarrow \quad \mathbb{P}\left(\left\{R(\widehat{g}_n) \geq R(g^\star) + \varepsilon\right\}\right) \leq \delta.$$